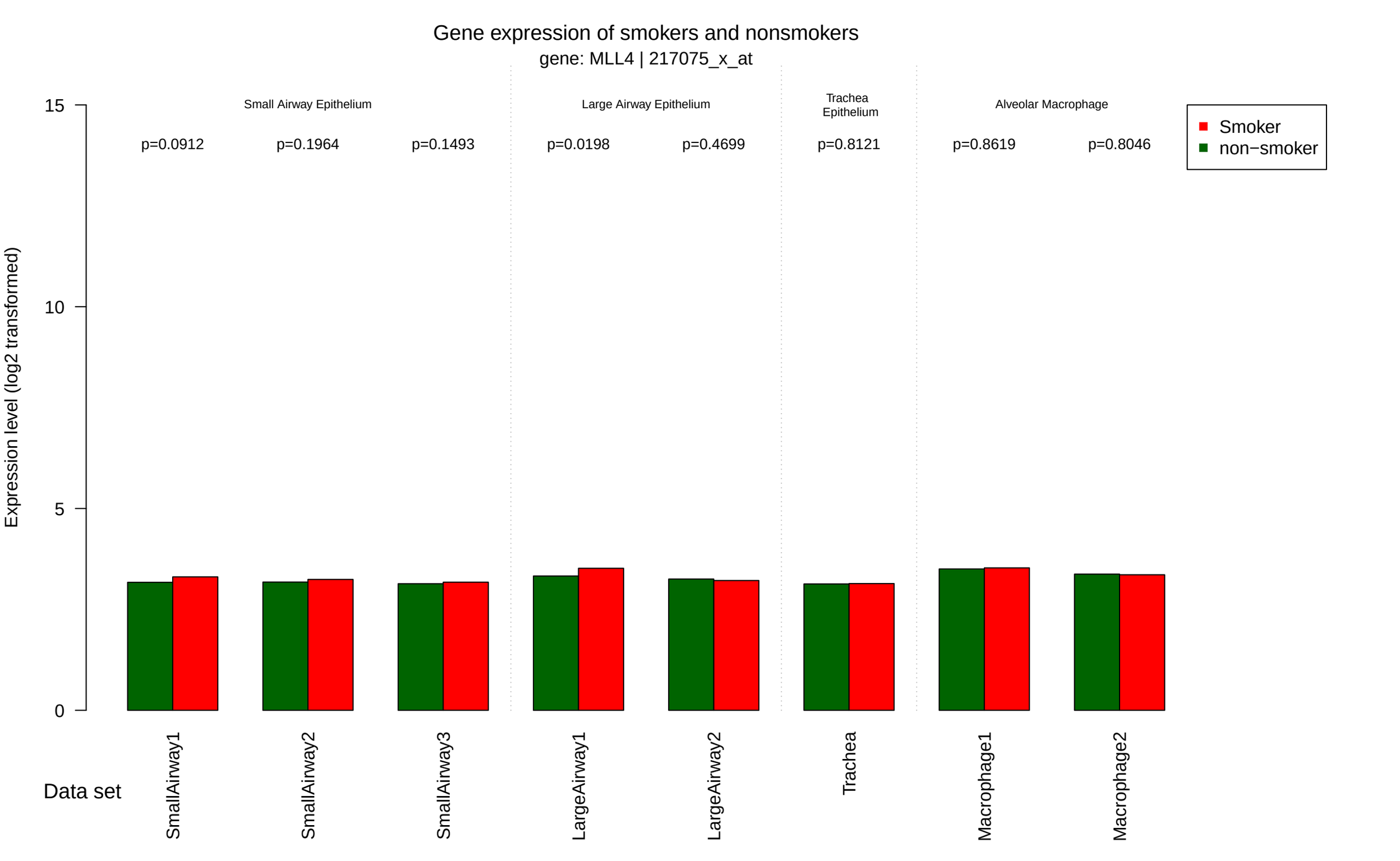
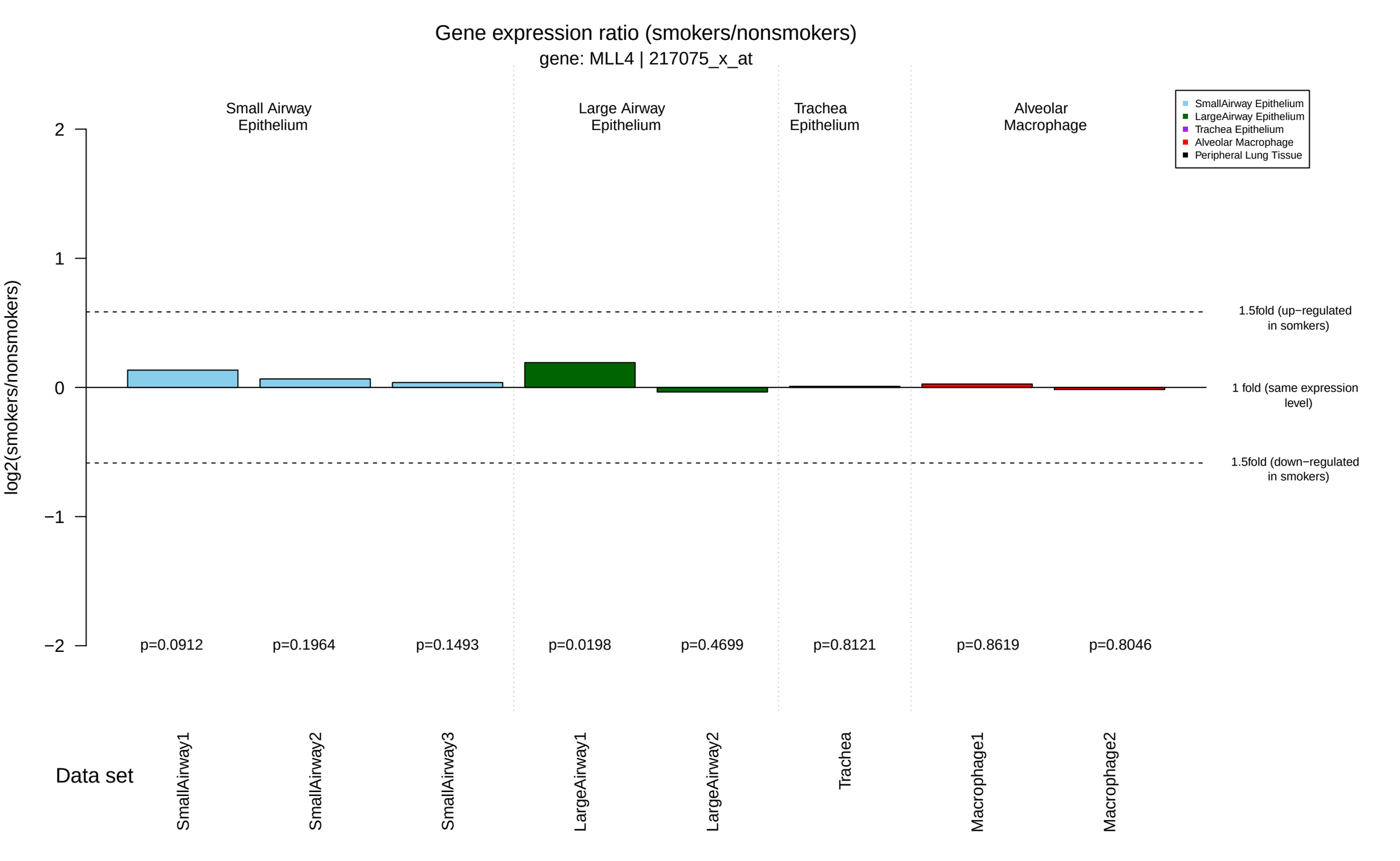
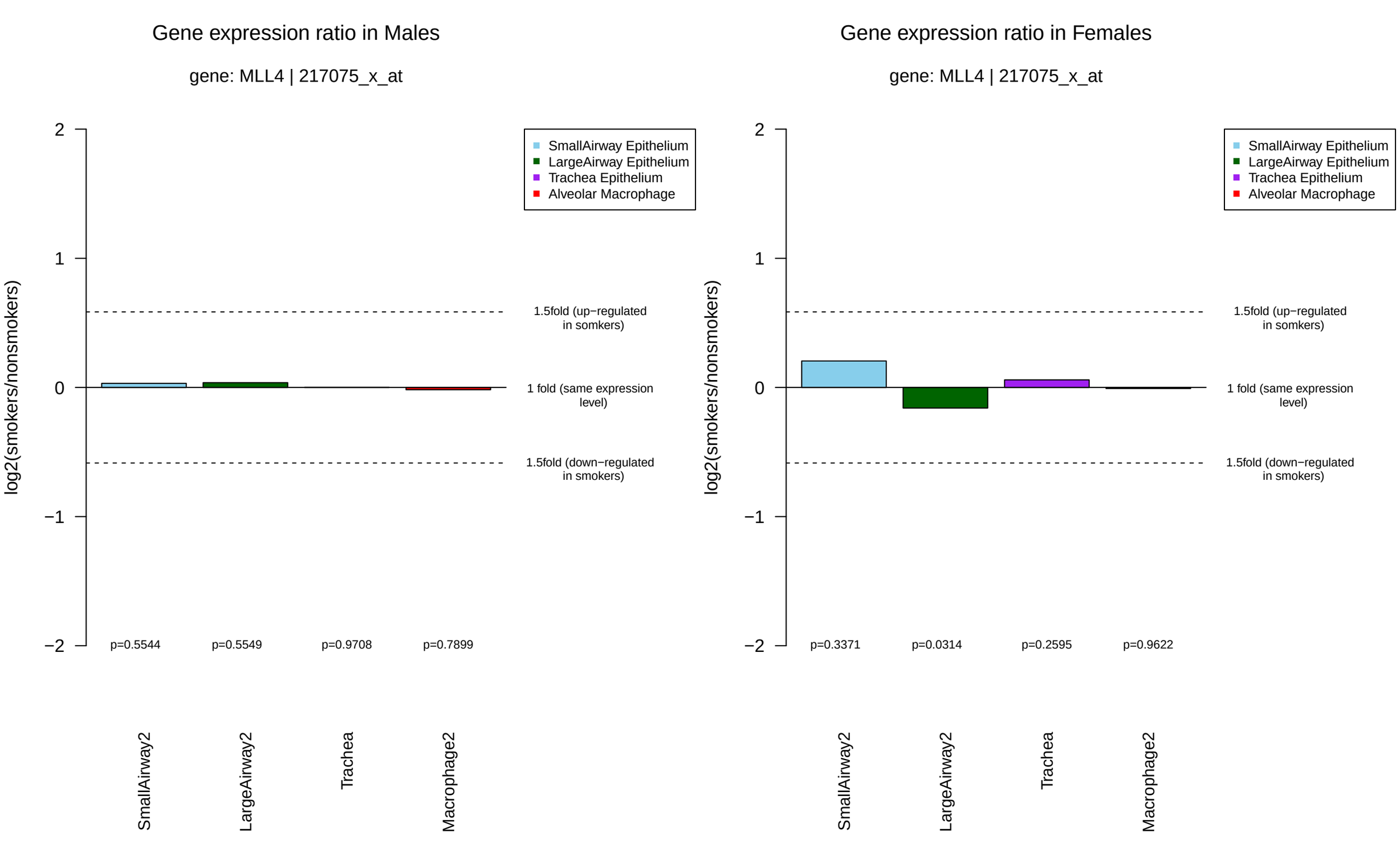
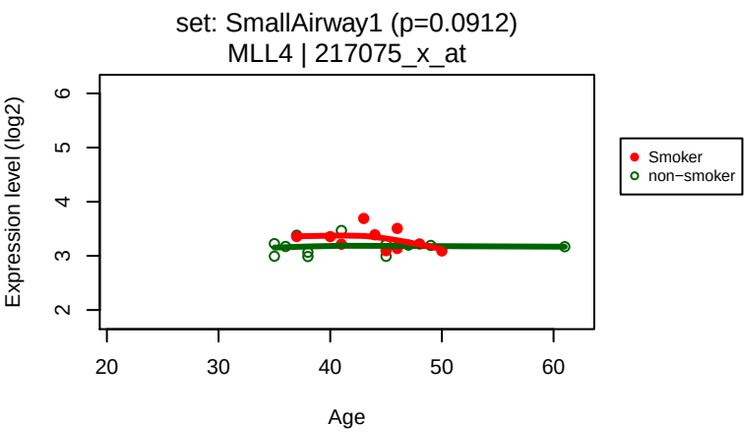
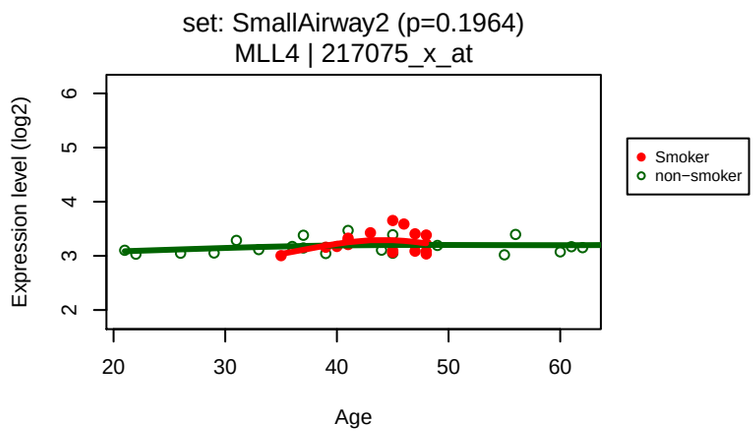
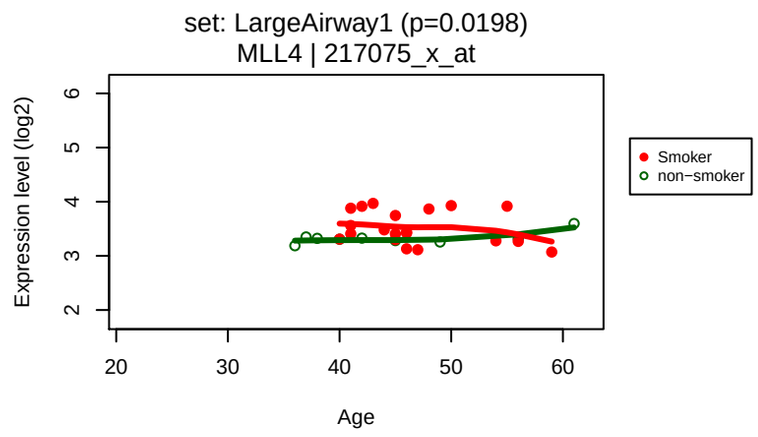

Gene expression of smokers and nonsmokers
gene: MLL4 | 217075_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0912
p=0.1964
p=0.1493
p=0.0198
p=0.4699
p=0.8121
p=0.8619
p=0.8046
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

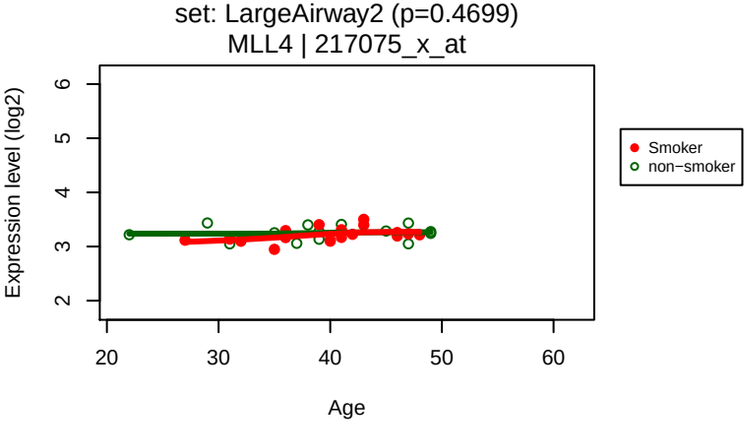

Gene expression ratio (smokers/nonsmokers)
gene: MLL4 | 217075_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0912
p=0.1964
p=0.1493
p=0.0198
p=0.4699
p=0.8121
p=0.8619
p=0.8046
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

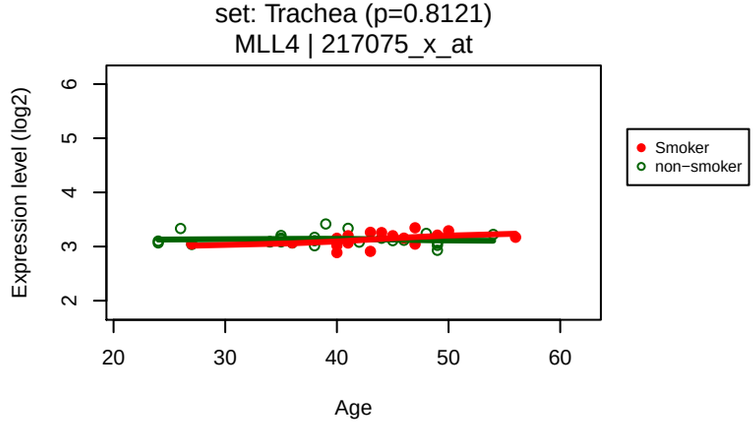

Gene expression ratio in Males
gene: MLL4 | 217075_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5544
p=0.5549
p=0.9708
p=0.7899
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MLL4 | 217075_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3371
p=0.0314
p=0.2595
p=0.9622
SmallAirway2
LargeAirway2
Trachea
Macrophage2

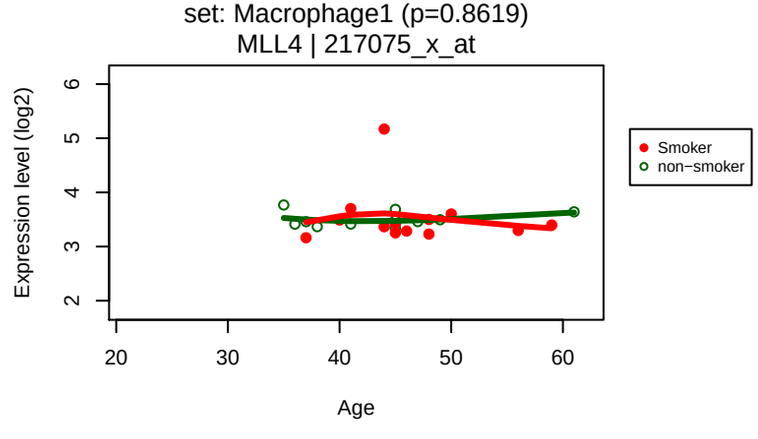

set: SmallAirway1 (p=0.0912)
MLL4 | 217075_x_at
Expression level (log2)
Age
Smoker
non−smoker

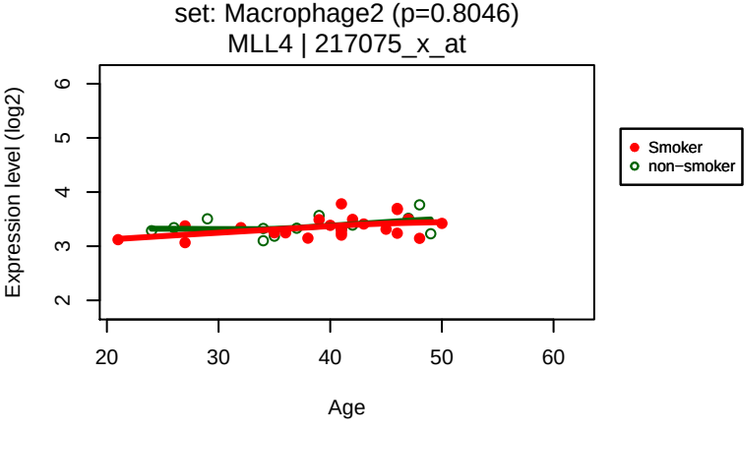

set: SmallAirway2 (p=0.1964)
MLL4 | 217075_x_at
Expression level (log2)
Age
Smoker
non−smoker

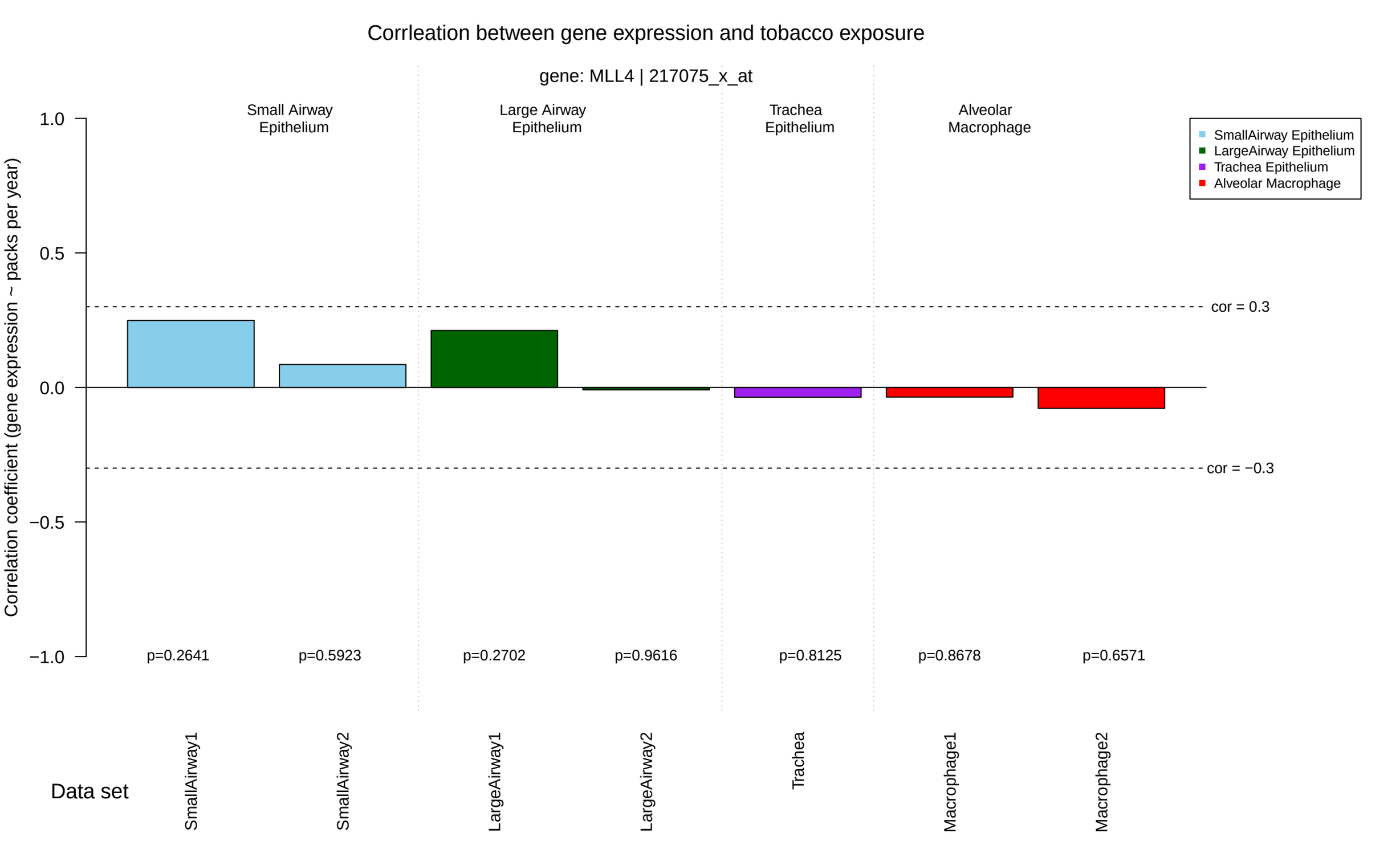

set: LargeAirway1 (p=0.0198)
MLL4 | 217075_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4699)
MLL4 | 217075_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8121)
MLL4 | 217075_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8619)
MLL4 | 217075_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8046)
MLL4 | 217075_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MLL4 | 217075_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2641
p=0.5923
p=0.2702
p=0.9616
p=0.8125
p=0.8678
p=0.6571
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2